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Bidirectional electrostatic discharge protection device

Abstract

Disclosed are bidirectional ESD protection devices capable of solving a breakdown voltage mismatch (BV mismatch) phenomenon while securing operation stability by providing a high breakdown voltage. The bidirectional ESD protection device includes an isolation region between an anode region and a cathode region, and the anode region and the cathode region may face each other with the isolation region in between. Accordingly, the bidirectional ESD protection device realizes high-voltage bidirectional characteristics without adding a mask, is minimized or reduced in size, and solves problems such as a breakdown voltage mismatch and instability.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
10276557	12/2018	Han et al.	N/A	N/A
2018/0082994	12/2017	Han et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
10-2018-0031175	12/2017	KR	N/A

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION

(1) The present application claims priority to Korean Patent Application No. 10-2022-0038501, filed Mar. 29, 2022, the entire contents of which are incorporated herein for all purposes by this reference.

BACKGROUND OF THE INVENTION

Field of the Invention

(2) The present disclosure concerns a bidirectional ESD protection device capable of solving a breakdown voltage mismatch (BV mismatch) phenomenon while securing operation stability (e.g., by having a high breakdown voltage). The present bidirectional ESD protection device includes an isolation region between an anode region and a cathode region, and the anode region and the cathode region face each other with the isolation region in between. Accordingly, the bidirectional ESD protection device realizes high-voltage bidirectional characteristics without adding a mask, is minimized or reduced in size, and solves problems such as a breakdown voltage mismatch and instability.

Description of the Related Art

(3) In general, devices such as a silicon controlled rectifier (SCR), a double-diffused MOS (DMOS) transistor, and a bipolar junction transistor (BJT) are used as high-voltage electrostatic discharge (ESD) protection devices.

(4) A conventional high-voltage bidirectional ESD protection device may be manufactured by connecting two ESD protection devices having a single high-voltage PNP structure in series (back to back series).

(5) For example, referring to FIG. 1, a conventional high-voltage bidirectional ESD protection device floats an emitter and a base (an isolation ring has the role of the base) through interconnection between two devices.

(6) In the conventional high-voltage bidirectional ESD protection device of FIG. 1, a reverse bias is

applied to a junction between a P well (P-type well) and a base of a P-type substrate (P-sub) when a cathode is subjected to positive zapping (e.g., when an ESD event results in a high voltage on the cathode). Accordingly, in the conventional high-voltage bidirectional ESD protection device, a high positive voltage may be applied to the base and the cathode when the cathode is subjected to positive zapping.

(7) In the conventional high-voltage bidirectional ESD protection device, a reverse bias is applied to a junction between a P well and a base of the cathode when the cathode is subjected to negative zapping (e.g., when an ESD event results in an unusually low negative voltage on the cathode). Accordingly, in the conventional high-voltage bidirectional ESD protection device, a high negative voltage may be applied to the cathode when the cathode is subjected to negative zapping.

(8) Accordingly, the conventional high-voltage bidirectional ESD protection device operates bidirectionally with respect to an anode connected to a ground potential (GND).

(9) However, the conventional high-voltage bidirectional ESD protection device has an isolation ring used as a base on both devices, so it is necessary to separate the two devices using the P-sub therebetween, resulting in a size/area increase.

(10) The conventional high-voltage bidirectional ESD protection device has therein the P well of the cathode and a parasitic PNP structure between the base and the P-sub, so in order for the ESD device to operate stably, it is necessary to make a breakdown voltage (BV) (e.g., the collector-emitter breakdown voltage, or BV_{cex}) of the parasitic PNP sufficiently greater than a breakdown voltage (e.g., BV_{cex}) of an ESD PNP. Herein, BV_{cex} refers to a breakdown voltage between a collector and an emitter while any bias (that is, generally, turn-on bias) is applied to a base.

(11) In particular, the conventional high-voltage bidirectional ESD protection device operates under a forward condition in which the collector is higher than the base when the parasitic PNP in the direction of the cathode is subjected to positive zapping, or operates under a forward condition in which the P-sub is higher than the base when the parasitic PNP is subjected to negative zapping.

(12) Therefore, in the conventional high-voltage bidirectional ESD protection device, the BV_{cex} or threshold voltage (e.g., V_{t1}) of the parasitic PNP should be designed to be sufficiently larger than the corresponding V_{t1} of the ESD PNP of the anode. Herein, V_{t1} is a voltage (V_c) right before a resistance arises between an emitter and a collector in a bipolar junction transistor (BJT) because of snapback.

(13) Fundamentally, the conventional high-voltage bidirectional ESD protection device should satisfy the above-described conditions in order to secure operation stability when the base of the ESD PNP shares the isolation ring. To this end, one should minimize the gain of the parasitic PNP and increase V_{t1} (SOA). Herein, V_{t1} (SOA) refers to a voltage current domain (that is, safe of area) that does not exceed V_{t1} according to a base voltage (V_{base}), wherein V_{t1} varies with the base voltage.

(14) The details described above are intended to help understanding of the background of the disclosure, and may include matters that are not publicly known or in the related art.

DOCUMENT OF RELATED ART

(15) Korean Patent Application Publication No. 10-2018-0031175.

SUMMARY OF THE INVENTION

(16) The present disclosure is to solve the problems in the related art described above.

(17) The present disclosure is directed to providing a bidirectional ESD protection device capable of solving a breakdown voltage mismatch (BV mismatch) phenomenon while securing operation stability and providing a breakdown voltage (BV) of 60 V or greater in bipolar ESD device manufactured in a bipolar-CMOS-DMOS (BCD) process.

(18) In addition, the present disclosure is directed to providing a bidirectional ESD protection device, in which a single bidirectional ESD protection device (e.g., a PNP type or an NPN type device) has a reduced area, a parasitic PNP including a base region in the device has an improved BV_{cex} to secure stability of ESD operation characteristics, and the bidirectional ESD protection

device reduces, minimizes or prevents a breakdown voltage variation (BV variation) and/or a left-right mismatch to secure a stable breakdown voltage.

(19) The present disclosure may be implemented by one or more embodiments having one or more of the following configurations to achieve the above-described objectives.

(20) According to one or more embodiments of the present disclosure, there is provided a bidirectional ESD protection device including a P-type epitaxial layer, an N-type buried layer in the epitaxial layer, a P well at or adjacent to an upper surface of the epitaxial layer (and optionally, close to a side of the epitaxial layer or the bidirectional ESD protection device); a plurality of N wells at or adjacent to the upper surface of the epitaxial layer and spaced apart from each other, a plurality of doped layers under the N wells, respectively, in the epitaxial layer; a plurality of connection N wells connected to the doped layers and the buried layer, in the epitaxial layer; a plurality of isolation P wells at or adjacent to the upper surface of the epitaxial layer and respectively between adjacent ones of the plurality of N wells; a plurality of isolation deep wells connected to the isolation P wells, respectively, in the epitaxial layer; and an anode region, a cathode region, and an isolation region between the anode region and the cathode region, wherein the isolation region comprises two adjacent ones of the plurality of N wells and the isolation P well between the two adjacent ones of the plurality of N wells, and the anode region and the cathode region comprise remaining ones of the plurality of N wells (other than the two adjacent ones of the N wells) and the remaining ones of the plurality of isolation P wells (other than the isolation P well in the isolation region) (e.g., on opposite sides of the isolation region).

(21) According to another embodiment of the present disclosure, in the bidirectional ESD protection device, the plurality of N wells may include a first N well at or adjacent to the upper surface of the epitaxial layer and spaced apart from the P well; a second N well at or adjacent to the upper surface of the epitaxial layer and spaced apart from the first N well; a third N well at or adjacent to the upper surface of the epitaxial layer and spaced apart from the second N well; a fourth N well at or adjacent to the upper surface of the epitaxial layer and spaced apart from the third N well; a fifth N well at or adjacent to the upper surface of the epitaxial layer and spaced apart from the fourth N well; and a sixth N well at or adjacent to the upper surface of the epitaxial layer and spaced apart from the fifth N well.

(22) According to still another embodiment of the present disclosure, in the bidirectional ESD protection device, the plurality of doped layers may include a first doped layer at least partially surrounding the first N well, in the epitaxial layer; a second doped layer at least partially surrounding the second N well, in the epitaxial layer; a third doped layer at least partially surrounding the third N well, in the epitaxial layer; a fourth doped layer at least partially surrounding the fourth N well, in the epitaxial layer; a fifth doped layer at least partially surrounding the fifth N well, in the epitaxial layer, and a sixth doped layer at least partially surrounding the sixth N well, in the epitaxial layer.

(23) According to still another embodiment of the present disclosure, in the bidirectional ESD protection device, the plurality of connection N wells may include a first connection N well connecting the first doped layer with the buried layer, in the epitaxial layer; a second connection N well connecting the second doped layer with the buried layer, in the epitaxial layer; a third connection N well connecting the third doped layer with the buried layer, in the epitaxial layer; a fourth connection N well connecting the fourth doped layer with the buried layer, in the epitaxial layer; a fifth connection N well connecting the fifth doped layer with the buried layer, in the epitaxial layer, and a sixth connection N well connecting the sixth doped layer with the buried layer, in the epitaxial layer.

(24) According to still another embodiment of the present disclosure, in the bidirectional ESD protection device, the plurality of isolation P wells may include a first isolation P well at or adjacent to the upper surface of the epitaxial layer, and between the first N well and the second N well; a second isolation P well at or adjacent to the upper surface of the epitaxial layer, and

between the second N well and the third N well; a third isolation P well at or adjacent to the upper surface of the epitaxial layer, and between the third N well and the fourth N well; a fourth isolation P well at or adjacent to the upper surface of the epitaxial layer, and between the fourth N well and the fifth N well; and a fifth isolation P well at or adjacent to the upper surface of the epitaxial layer, and between the fifth N well and the sixth N well.

(25) According to still another embodiment of the present disclosure, in the bidirectional ESD protection device, the plurality of isolation deep wells may include a first isolation deep well connected to the first isolation P well, in the epitaxial layer, a second isolation deep well connected to the second isolation P well, in the epitaxial layer; a third isolation deep well connected to the third isolation P well, in the epitaxial layer, a fourth isolation deep well connected to the fourth isolation P well, in the epitaxial layer; and a fifth isolation deep well connected to the fifth isolation P well, in the epitaxial layer.

(26) According to still another embodiment of the present disclosure, in the bidirectional ESD protection device, the isolation region may comprise the third N well, the fourth N well, and the third isolation P well, the anode region may comprise the first N well, the second N well, the first isolation P well, and the second isolation P well (e.g., on one side of the isolation region), and the cathode region may comprise the fifth N well, the sixth N well, the fourth isolation P well, and the fifth isolation P well on an opposite side of the isolation region. The first N well, the second N well, the first isolation P well, and the second isolation P well may face the fifth N well, the sixth N well, the fourth isolation P well, and the fifth isolation P well, with the isolation region in the center.

(27) According to still another embodiment of the present disclosure, there is provided a bidirectional ESD protection device including a P-type epitaxial layer, an N-type buried layer in the epitaxial layer, a P well at or adjacent to an upper surface of the epitaxial layer (and optionally, close to a side of the epitaxial layer or the bidirectional ESD protection device); a plurality of N wells at or adjacent to the upper surface of the epitaxial layer and spaced apart from each other; a plurality of doped layers under the N wells, respectively, in the epitaxial layer, a plurality of connection N wells connected to the doped layers and the buried layer, in the epitaxial layer; a plurality of isolation P wells at or adjacent to the upper surface of the epitaxial layer and respectively between adjacent ones of the plurality of N wells; and a plurality of isolation deep wells connected to the isolation P wells, respectively, in the epitaxial layer; and an anode region, a cathode region, and an isolation region between the anode region and the cathode region, wherein the isolation region comprises two adjacent ones of the plurality of N wells and the isolation P well between the two adjacent ones of the N wells. The anode region and the cathode region may face each other with the isolation region in a center, and the anode region and the cathode region may be on opposite sides of the isolation region.

(28) According to still another embodiment of the present disclosure, in the bidirectional ESD protection device, the plurality of N wells may include a first N well at or adjacent to the upper surface of the epitaxial layer and spaced apart from the P well; a second N well at or adjacent to the upper surface of the epitaxial layer and spaced apart from the first N well; a third N well at or adjacent to the upper surface of the epitaxial layer and spaced apart from the second N well; a fourth N well at or adjacent to the upper surface of the epitaxial layer and spaced apart from the third N well; a fifth N well at or adjacent to the upper surface of the epitaxial layer and spaced apart from the fourth N well; and a sixth N well at or adjacent to the upper surface of the epitaxial layer and spaced apart from the fifth N well.

(29) According to still another embodiment of the present disclosure, in the bidirectional ESD protection device, the plurality of doped layers may include a first doped layer at least partially surrounding the first N well, in the epitaxial layer; a second doped layer at least partially surrounding the second N well, in the epitaxial layer; a third doped layer at least partially surrounding the third N well, in the epitaxial layer; a fourth doped layer at least partially

surrounding the fourth N well, in the epitaxial layer, a fifth doped layer at least partially surrounding the fifth N well, in the epitaxial layer, and a sixth doped layer at least partially surrounding the sixth N well, in the epitaxial layer.

(30) According to still another embodiment of the present disclosure, in the bidirectional ESD protection device, the plurality of connection N wells may include a first connection N well under the first doped layer connecting the first doped layer with the buried layer, in the epitaxial layer; a second connection N well under the third doped layer connecting the third doped layer with the buried layer, in the epitaxial layer; a third connection N well under the fourth doped layer connecting the fourth doped layer with the buried layer, in the epitaxial layer; and a fourth connection N well under the sixth doped layer connecting the sixth doped layer with the buried layer, in the epitaxial layer.

(31) According to still another embodiment of the present disclosure, in the bidirectional ESD protection device, the plurality of isolation P wells may include a first isolation P well at or adjacent to the upper surface of the epitaxial layer, and between the first N well and the second N well; a second isolation P well at or adjacent to the upper surface of the epitaxial layer, and between the second N well and the third N well; a third isolation P well at or adjacent to the upper surface of the epitaxial layer, and between the third N well and the fourth N well; a fourth isolation P well at or adjacent to the upper surface of the epitaxial layer, and between the fourth N well and the fifth N well; and a fifth isolation P well at or adjacent to the upper surface of the epitaxial layer, and between the fifth N well and the sixth N well.

(32) According to still another embodiment of the present disclosure, in the bidirectional ESD protection device, the plurality of isolation deep wells may include a first isolation deep well connected to the first isolation P well, in the epitaxial layer, a second isolation deep well connected to the second isolation P well, in the epitaxial layer; a third isolation deep well connected to the third isolation P well, in the epitaxial layer, a fourth isolation deep well connected to the fourth isolation P well, in the epitaxial layer; and a fifth isolation deep well connected to the fifth isolation P well, in the epitaxial layer.

(33) According to still another embodiment of the present disclosure, in the bidirectional ESD protection device, the isolation region may comprise the third N well, the fourth N well, and the third isolation P well, the anode region may comprise the first N well, the second N well, the first isolation P well, and the second isolation P well (e.g., on one side of the isolation region), and the fifth N well, the sixth N well, the fourth isolation P well, and the cathode region may comprise the fifth isolation P well (e.g., on an opposite side of the isolation region). The first N well, the second N well, the first isolation P well, and the second isolation P well may face the fifth N well, the sixth N well, the fourth isolation P well, and the fifth isolation P well, with the isolation region in the center.

(34) According to still another embodiment of the present disclosure, there is provided a bidirectional ESD protection device including a P-type epitaxial layer, an N-type buried layer in the epitaxial layer, a P well at or adjacent to an upper surface of the epitaxial layer (and optionally, close to a side of the epitaxial layer or the bidirectional ESD protection device); a plurality of N wells at or adjacent to the upper surface of the epitaxial layer and spaced apart from each other; a plurality of doped layers under the N wells, respectively, in the epitaxial layer, a plurality of connection N wells connected to the doped layers and the buried layer, in the epitaxial layer; a plurality of isolation P wells at or adjacent to the upper surface of the epitaxial layer and respectively between each of the plurality of N wells; a plurality of isolation deep wells connected to the isolation P wells, respectively, in the epitaxial layer, and an anode region, a cathode region, and an isolation region between the anode region and the cathode region, wherein the isolation region comprises one of the plurality of N wells, and the anode region and the cathode region comprise remaining ones of the plurality of N wells (other than the N well in the isolation region) and the plurality of isolation P wells. The remaining ones of the plurality of N wells and the

plurality of isolation P wells may face each other with the isolation region in the center, and the anode region and the cathode region may be on opposite sides of the isolation region.

(35) According to still another embodiment of the present disclosure, in the bidirectional ESD protection device, the plurality of N wells may include a first N well at or adjacent to the upper surface of the epitaxial layer and spaced apart from the P well; a second N well at or adjacent to the upper surface of the epitaxial layer and spaced apart from the first N well; a third N well at or adjacent to the upper surface of the epitaxial layer and spaced apart from the second N well; a fourth N well at or adjacent to the upper surface of the epitaxial layer and spaced apart from the third N well; and a fifth N well at or adjacent to the upper surface of the epitaxial layer and spaced apart from the fourth N well, and the plurality of isolation P wells may include a first isolation P well at or adjacent to the upper surface of the epitaxial layer, and between the first N well and the second N well; a second isolation P well at or adjacent to the upper surface of the epitaxial layer, and between the second N well and the third N well; a third isolation P well at or adjacent to the upper surface of the epitaxial layer, and between the third N well and the fourth N well; and a fourth isolation P well at or adjacent to the upper surface of the epitaxial layer, and between the fourth N well and the fifth N well.

(36) According to still another embodiment of the present disclosure, in the bidirectional ESD protection device, the plurality of doped layers may include a first doped layer at least partially surrounding the first N well, in the epitaxial layer; a second doped layer at least partially surrounding the second N well, in the epitaxial layer; a third doped layer at least partially surrounding the third N well, in the epitaxial layer; a fourth doped layer at least partially surrounding the fourth N well, in the epitaxial layer; and a fifth doped layer at least partially surrounding the fifth N well, in the epitaxial layer.

(37) According to still another embodiment of the present disclosure, in the bidirectional ESD protection device, the plurality of connection N wells may include a first connection N well connecting the first doped layer with the buried layer, in the epitaxial layer; a second connection N well connecting the third doped layer with the buried layer, in the epitaxial layer; and a third connection N well connecting the fifth doped layer with the buried layer, in the epitaxial layer.

(38) According to still another embodiment of the present disclosure, in the bidirectional ESD protection device, the plurality of isolation deep wells may include a first isolation deep well connected to the first isolation P well under the first isolation P well, in the epitaxial layer; a second isolation deep well connected to the second isolation P well under the second isolation P well, in the epitaxial layer, a third isolation deep well connected to the third isolation P well under the third isolation P well, in the epitaxial layer; and a fourth isolation deep well connected to the fourth isolation P well under the fourth isolation P well, in the epitaxial layer.

(39) According to still another embodiment of the present disclosure, in the bidirectional ESD protection device, the isolation region may comprise the third N well between the anode region and the cathode region, the anode region may comprise the first N well, the second N well, the first isolation P well, and the second isolation P well on one side of the isolation region, and the cathode region may comprise the fourth N well, the fifth N well, the third isolation P well, and the fourth isolation P well on an opposite side of the isolation region. The first N well, the second N well, the first isolation P well, and the second isolation P well may face the fourth N well, the fifth N well, the third isolation P well, and the fourth isolation P well, with the isolation region in the center.

(40) According to the above configurations, the present disclosure has the following effects.

(41) According to the present disclosure, the bidirectional ESD protection device can realize high-voltage bidirectional characteristics without adding a mask.

(42) In addition, since a mask is not added, the bidirectional ESD protection device can realize bidirectional characteristics and can be minimized or reduced in size.

(43) In addition, the bidirectional ESD protection device can be designed to have the same breakdown voltage or an asymmetric breakdown voltage according to an operation condition by

realizing a PNP structure in which the anode region and the cathode region face each other with the isolation region in between (e.g., are on opposite sides of the isolation region).

(44) In addition, the bidirectional ESD protection device can solve the problem in the related art, an unstable breakdown voltage, by providing a high-voltage region with a stable breakdown voltage.

(45) It is noted that, although not explicitly described under this section, an advantageous effect and a tentative advantageous effect that are expected from technical features of the present disclosure are regarded as being described in the specification of the present disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The above and other objectives, features, and other advantages of the present disclosure will be more clearly understood from the following detailed description when taken in conjunction with the accompanying drawings, in which:

(2) FIG. 1 is a diagram illustrating a conventional bidirectional ESD protection device;

(3) FIGS. 2 and 3 are diagrams illustrating a bidirectional ESD protection device according to a first exemplary embodiment of the present disclosure;

(4) FIGS. 4 and 5 are diagrams illustrating a bidirectional ESD protection device according to a second exemplary embodiment of the present disclosure;

(5) FIGS. 6 and 7 are diagrams illustrating a bidirectional ESD protection device according to a third exemplary embodiment of the present disclosure; and

(6) FIG. 8 is a diagram illustrating comparison between the bidirectional ESD protection devices according to the exemplary embodiments of the present disclosure and the conventional bidirectional ESD protection device.

DETAILED DESCRIPTION OF THE INVENTION

(7) Hereinafter, embodiments of the present disclosure will be described in detail with reference to the accompanying drawings. It is noted that the embodiments of the present disclosure may be changed to a variety of other or different embodiments. The scope of the present disclosure should not be interpreted as being limited to the embodiments described hereinbelow, but should be interpreted on the basis of the descriptions in the appended claims. In addition, the embodiments of the present disclosure are provided for reference in order to fully describe the disclosure for those skilled in the art.

(8) Unless otherwise mentioned in context, a singular noun or a singular noun phrase may have a plural meaning through the present specification. The terms “comprise” and/or “comprising” that are used in the present specification are intended to indicate that a shape, a number, a step, an operation, a member, an element, and/or a group thereof is/are present, and therefore do not to preclude the presence or addition of one or more other shapes, numbers, steps, operations, members, elements, and/or groups thereof.

(9) It should be noted that, in a case where one element (or layer) is described as being on another element (or layer), this means that the one element may be directly on the other element or that one or more third elements (or layers) may be therebetween. In addition, in the case where one element is described as being directly on the other element, no third element(s) is therebetween. In addition, positioning on a “top”, “upper portion”, or “lower portion” of one element, positioning “above” or “below” one element, or positioning on “one lateral side”, or a “lateral surface” of one element means a relative positional relationship between the elements.

(10) In the embodiments described below, a first conductivity type may be P-type, and a second conductivity type may be N-type, for example, but are not necessarily limited thereto.

(11) FIGS. 2 and 3 are diagrams illustrating a bidirectional ESD protection device according to a first exemplary embodiment of the present disclosure.

(12) Hereinafter, a bidirectional ESD protection device according to embodiments of the present disclosure will be described in detail with reference to the accompanying drawings.

(13) Referring to FIGS. 2 and 3, the bidirectional ESD protection device **100** according to the first exemplary embodiment of the present disclosure includes an epitaxial layer **110**, a buried layer **120**, a P well **130**, a first N well **141**, a second N well **142**, a third N well **143**, a fourth N well **144**, a fifth N well **145**, a sixth N well **146**, a first doped layer **151**, a second doped layer **152**, a third doped layer **153**, a fourth doped layer **154**, a fifth doped layer **155**, a sixth doped layer **156**, a first connection N well **161**, a second connection N well **162**, a third connection N well **163**, a fourth connection N well **164**, a fifth connection N well **165**, a sixth connection N well **166**, a first isolation P well **171**, a second isolation P well **172**, a third isolation P well **173**, a fourth isolation P well **174**, a fifth isolation P well **175**, a first isolation deep well **181**, a second isolation deep well **182**, a third isolation deep well **183**, a fourth isolation deep well **184**, and a fifth isolation deep well **185**.

(14) The epitaxial layer **110** is on one surface of a semiconductor substrate **195**. The epitaxial layer **110** has the first conductivity type, and may comprise elemental silicon and be formed by an epitaxial growth process. For example, the epitaxial layer **110** may further comprise one or more dopants and be doped by adding the dopant(s) during the epitaxial growth process. The epitaxial layer may also be doped by ion implantation after the epitaxial layer is formed.

(15) The buried layer **120** is in the epitaxial layer **110** and has a predetermined width. The buried layer **120** may be close to a lower surface of the epitaxial layer **110**, inside the epitaxial layer **110**. The buried layer **120** is connected to lower surfaces of the first connection N well **161** to the sixth connection N well **166**. The buried layer **120** comprises antimony, a material of the second conductivity type, for example.

(16) The P well **130** is in the epitaxial layer **110**, at or adjacent to an upper surface of the epitaxial layer **110**. The P well **130** may be at one outermost side or border of the device **100** (e.g., close to the left or right side of the epitaxial layer **110** in the drawing). The P well **130** may be formed by injecting high-concentration first-conductivity-type impurities into the surface of the epitaxial layer **110** through a photolithographically-patterned mask. The P well **130** is connected to (e.g., in ohmic contact with) a P-sub terminal and/or electrode.

(17) The first N well **141** to the sixth N well **146** are in the epitaxial layer **110**, at or adjacent to the upper surface of the epitaxial layer **110**. The first N well **141** to the sixth N well **146** are between the right side of the epitaxial layer **110** and the P well **130** in the drawing, and spaced apart from each other by a predetermined distance.

(18) The first N well **141** is adjacent to the P well **130** and connected to an M1 terminal and/or electrode. The second N well **142** is adjacent to the first N well **141** and connected to an emitter terminal and/or electrode. That is, a P+ region adjacent to the second N well **142** may be connected to the anode collector terminal and/or electrode, the second N well **142** is connected to the anode emitter terminal and/or electrode.

(19) The third N well **143** is adjacent to the second N well **142** and connected to a base terminal and/or electrode.

(20) The fourth N well **144** is adjacent to the third N well **143** and connected to the base terminal and/or electrode.

(21) The fifth N well **145** is adjacent to the fourth N well **144** and connected to an emitter terminal and/or electrode. That is, a P+ region adjacent to the fifth N well **145** may be connected to the cathode collector terminal and/or electrode, the fifth N well **145** is connected to the cathode emitter terminal and/or electrode. The sixth N well **146** is at or adjacent to the fifth N well **145** and connected to a base terminal and/or electrode. Thus, the second N well **142** may be between the first N well **141** and the third N well **143**, the third N well **143** may be between the second N well **142** and the fourth N well **144**, the fourth N well **144** may be between the third N well **143** and the fifth N well **145**, and the fifth N well **145** may be between the fourth N well **144** and the sixth N

well **146**.

(22) The first doped layer **151** to the sixth doped layer **156** are in the epitaxial layer **110**, and under the first N well **141** to the sixth N well **146**.

(23) The first doped layer **151** to the sixth doped layer **156** may overlap with parts of the first N well **141** to the sixth N well **146**, respectively. That is, the first doped layer **151** is under the first N well **141**, and may overlap with the first N well **141**. The second doped layer **152** is under the second N well **142**, and may overlap with the second N well **142**. The third doped layer **153** is under the third N well **143**, and may overlap with the third N well **143**. The fourth doped layer **154** is under the fourth N well **144**, and may overlap with the fourth N well **144**. The fifth doped layer **155** is under the fifth N well **145**, and may overlap with the fifth N well **145**. The sixth doped layer **156** is under the sixth N well **146**, and may overlap with the sixth N well **146**.

(24) The first connection N well **161** to the sixth connection N well **166** are in the epitaxial layer **110**, and may be under the first doped layer **151** to the sixth doped layer **156**, respectively.

(25) The first connection N well **161** to the sixth connection N well **166** connect the buried layer **120** to the first doped layer **151** to the sixth doped layer **156** or lower surfaces thereof, respectively. That is, in the drawing, an upper portion of the first connection N well **161** is connected to the first doped layer **151**, and a lower portion of the first connection N well **161** is connected to the buried layer **120**. In the drawing, an upper portion of the second connection N well **162** is connected to the second doped layer **152**, and a lower portion of the second connection N well **162** is connected to the buried layer **120**. In the drawing, an upper portion of the third connection N well **163** is connected to the third doped layer **153**, and a lower portion of the third connection N well **163** is connected to the buried layer **120**. In the drawing, an upper portion of the fourth connection N well **164** is connected to the fourth doped layer **154**, and a lower portion of the fourth connection N well **164** is connected to the buried layer **120**. In the drawing, an upper portion of the fifth connection N well **165** is connected to the fifth doped layer **155**, and a lower portion of the fifth connection N well **165** is connected to the buried layer **120**. In the drawing, an upper portion of the sixth connection N well **166** is connected to the sixth doped layer **156**, and a lower portion of the sixth connection N well **166** is connected to the buried layer **120**.

(26) The first isolation P well **171** to the fifth isolation P well **175** are in the epitaxial layer **110**, and at or adjacent to the upper surface of the epitaxial layer **110**. The first isolation P well **171** to the fifth isolation P well **175** are spaced apart from each other by a predetermined distance, and are respectively between adjacent ones of the N wells.

(27) The first isolation P well **171** is between the first N well **141** and the second N well **142** and connected to a first (e.g., anode) collector terminal and/or electrode. The second isolation P well **172** is between the second N well **142** and the third N well **143** and connected to the first collector terminal and/or electrode. The third isolation P well **173** is between the third N well **143** and the fourth N well **144** and connected to the base terminal and/or electrode. The fourth isolation P well **174** is between the fourth N well **144** and the fifth N well **145** and connected to a second (e.g., cathode) collector terminal and/or electrode. The fifth isolation P well **175** is between the fifth N well **145** and the sixth N well **146** and connected to the second collector terminal and/or electrode.

(28) The first isolation deep well **181** to the fifth isolation deep well **185** are in the epitaxial layer **110**, and under the first isolation P well **171** to the fifth isolation P well **175**. The first isolation deep well **181** is under the first isolation P well **171**, and connected to a lower portion of the first isolation P well **171**. The second isolation deep well **182** is under the second isolation P well **172**, and connected to a lower portion of the second isolation P well **172**. The third isolation deep well **183** is under the third isolation P well **173**, and connected to a lower portion of the third isolation P well **173**. The fourth isolation deep well **184** is under the fourth isolation P well **174**, and connected to a lower portion of the fourth isolation P well **174**. The fifth isolation deep well **185** is under the fifth isolation P well **175**, and connected to a lower portion of the fifth isolation P well **175**.

(29) The third N well **143**, the third isolation P well **173**, and the fourth N well **144** connected to

the base terminal and/or electrode in the center of the drawing constitute an isolation region **190**. Herein, the isolation region **190** serves as a region for isolation between an anode on the left side of the drawing and a cathode on the right side of the drawing. The isolation region **190** includes the third doped layer **153** and the third connection N well **163** connected to the third N well **143**, the third isolation deep well **183** connected to the third isolation P well **173**, the fourth doped layer **154** and the fourth connection N well **164** connected to the fourth N well **144**, and part of the buried layer **120**.

(30) Herein, the P-sub terminal and/or electrode and the collector terminals and/or electrodes on the left side of the drawing constitute the anode, and the collector terminals and/or electrodes on the right side of the drawing constitute the cathode. The device **100** may function as a type or kind of bidirectional diode, in which only voltages greater than a first positive threshold and voltages less than a second negative threshold pass through the device.

(31) The P well **130**, the first isolation P well **171**, the first isolation deep well **181**, the second isolation P well **172**, and the second isolation deep well **182** constitute an anode region. Herein, the anode region may include the first N well **141**, the first doped layer **151**, the first connection N well **161**, the second N well **142**, the second doped layer **152**, and the second connection N well **162**.

(32) The fourth isolation P well **174**, the fourth isolation deep well **184**, the fifth isolation P well **175**, and the fifth isolation deep well **185** constitute a cathode region. Herein, the cathode region may include the fifth N well **145**, the fifth doped layer **155**, the fifth connection N well **165**, the sixth N well **146**, the sixth doped layer **156**, and the sixth connection N well **166**.

(33) The isolation region **190** constitutes an isolation structure having a very large BV_{cex} between the anode region and the cathode region. That is, the BV_{cex} of the isolation region **190** has a value greater than the breakdown voltages (BV₂ @ cathode positive and BV₁ @ cathode positive) when the cathode is subjected to positive zapping. Herein, the breakdown voltage BV₂ @ cathode positive has a value greater than the breakdown voltage BV₁ @ cathode positive, and has the relation below. Herein, the BV_{cex} is the breakdown voltage between the collector connected to the cathode and the collector connected to the anode while any bias (that is, generally, turn-on bias) is applied to the base (i.e., the isolation region between the anode and the cathode):

$BV_{cex} > BV_2 @ \text{cathode positive} > BV_1 @ \text{cathode positive}$

(34) In addition, the BV_{cex} of the isolation region **190** has a value greater than breakdown voltages (BV₃ @ cathode negative and BV₄ @ cathode negative) when the cathode is subjected to negative zapping. Herein, the breakdown voltage BV₄ @ cathode negative has a value greater than the breakdown voltage BV₃ @ cathode negative, and has the relation below:

$BV_{cex} > BV_4 @ \text{Cathode Negative} > BV_3 @ \text{Cathode Negative}$

(35) In addition, the isolation P well **130** and the base (that is, a PNP device) in the anode region or the cathode region may comprise a multi-finger device or have a multi-finger layout.

(36) Accordingly, the bidirectional ESD protection device **100** according to the first exemplary embodiment of the present disclosure can secure the same breakdown voltages (BVs) in positive zapping and negative zapping when the PNP transistors in the anode region and the cathode region are the same.

(37) FIGS. **4** and **5** are diagrams illustrating a bidirectional ESD protection device according to a second exemplary embodiment of the present disclosure.

(38) Referring to FIGS. **4** and **5**, the bidirectional ESD protection device **200** according to the second exemplary embodiment of the present disclosure includes an epitaxial layer **210**, a buried layer **220**, a P well **230**, a first N well **241**, a second N well **242**, a third N well **243**, a fourth N well **244**, a fifth N well **245**, a sixth N well **246**, a first doped layer **251**, a second doped layer **252**, a third doped layer **253**, a fourth doped layer **254**, a fifth doped layer **255**, a sixth doped layer **256**, a first connection N well **261**, a second connection N well **262**, a third connection N well **263**, a fourth connection N well **264**, a first isolation P well **271**, a second isolation P well **272**, a third isolation P well **273**, a fourth isolation P well **274**, a fifth isolation P well **275**, a first isolation deep

well **281**, a second isolation deep well **282**, a third isolation deep well **283**, a fourth isolation deep well **284**, and a fifth isolation deep well **285**.

(39) The epitaxial layer **210** is on one surface of a semiconductor substrate **295**. The epitaxial layer **210** has a P-type conductivity, and may comprise elemental silicon and be formed by an epitaxial growth process. For example, the epitaxial layer **210** may further comprise one or more dopants and be doped by adding the dopant(s) during the epitaxial growth process. The epitaxial layer may also be doped by ion implantation after the epitaxial layer is formed.

(40) The buried layer **220** is in the epitaxial layer **210** and has a predetermined width. The buried layer **220** may be close to a lower surface of the epitaxial layer **210**, inside the epitaxial layer **210**. The buried layer **220** is connected to lower surfaces of the first connection N well **261** to the fourth connection N well **264**. The buried layer **220** comprises antimony, a material of the second conductivity type, for example.

(41) The P well **230** is in the epitaxial layer **210**, and at or adjacent to an upper surface of the epitaxial layer **210**. The P well **230** may be at one outermost side or border of the device **200** (e.g., close to the left or right side of the epitaxial layer **210** in the drawing). The P well **230** may be formed by injecting high-concentration first-conductivity-type impurities into the surface of the epitaxial layer **210** through a photolithographically-patterned mask. The P well **230** is connected (e.g., in ohmic contact with) to a P-sub terminal and/or electrode.

(42) The first N well **241** to the sixth N well **246** are in the epitaxial layer **210**, and at or adjacent to the upper surface of the epitaxial layer **210**. The first N well **241** to the sixth N well **246** are between the right side of the epitaxial layer **210** and the P well **230** in the drawing, and spaced apart from each other by a predetermined distance.

(43) The first N well **241** is at or adjacent to the P well **230** and connected to an M1 terminal and/or electrode. The second N well **242** is at or adjacent to the first N well **241** and connected to an emitter terminal and/or electrode. That is, a P⁺ region adjacent to the second N well **242** may be connected to the anode collector terminal and/or electrode, and the second N well **242** is connected to the anode emitter terminal and/or electrode.

(44) The third N well **243** is adjacent to the second N well **242** and connected to a base terminal and/or electrode.

(45) The fourth N well **244** is adjacent to the third N well **243** and connected to the base terminal and/or electrode.

(46) The fifth N well **245** is adjacent to the fourth N well **244** and connected to an emitter terminal and/or electrode. That is, a P⁺ region adjacent to the fifth N well **245** is connected to the cathode collector terminal and/or electrode, and the fifth N well **245** is connected to the cathode emitter terminal and/or electrode.

(47) The sixth N well **246** is adjacent to the fifth N well **245** and connected to a base terminal and/or electrode. Thus, the second N well **242** is between the first N well **241** and the third N well **243**, the third N well **243** is between the second N well **242** and the fourth N well **244**, the fourth N well **244** is between the third N well **243** and the fifth N well **245**, and the fifth N well **245** is between the fourth N well **244** and the sixth N well **246**.

(48) The first doped layer **251** to the sixth doped layer **256** are in the epitaxial layer **210**, and under the first N well **241** to the sixth N well **246**.

(49) The first doped layer **251** to the sixth doped layer **256** may overlap with parts of the first N well **241** to the sixth N well **246**, respectively. That is, the first doped layer **251** is under the first N well **241**, and may overlap with the first N well **241**. The second doped layer **252** is under the second N well **242**, and may overlap with the second N well **242**. The third doped layer **253** is under the third N well **243**, and may overlap with the third N well **243**. The fourth doped layer **254** is under the fourth N well **244**, and may overlap with the fourth N well **244**. The fifth doped layer **255** is under the fifth N well **245**, and may overlap with the fifth N well **245**. The sixth doped layer **256** is under the sixth N well **246**, and may overlap with the sixth N well **246**.

(50) The first connection N well **261** to the fourth connection N well **264** are in the epitaxial layer **210**. The first connection N well **261** to the fourth connection N well **264** may connect the buried layer **220** to the first doped layer **251**, the third doped layer **253**, the fourth doped layer **254**, and the sixth doped layer **256** or lower surfaces thereof, respectively. That is, in the drawing, an upper portion of the first connection N well **261** is connected to the first doped layer **251**, and a lower portion of the first connection N well **261** is connected to the buried layer **220**. In the drawing, an upper portion of the second connection N well **262** is connected to the third doped layer **253**, and a lower portion of the second connection N well **262** is connected to the buried layer **220**. In the drawing, an upper portion of the third connection N well **263** is connected to the fourth doped layer **254**, and a lower portion of the third connection N well **263** is connected to the buried layer **220**. In the drawing, an upper portion of the fourth connection N well **264** is connected to the sixth doped layer **256**, and a lower portion of the fourth connection N well **264** is connected to the buried layer **220**.

(51) The first isolation P well **271** to the fifth isolation P well **275** are in the epitaxial layer **210**, and at or adjacent to the upper surface of the epitaxial layer **210**. The first isolation P well **271** to the fifth isolation P well **275** are spaced apart from each other by a predetermined distance, and are respectively between adjacent ones of the N wells.

(52) The first isolation P well **271** is between the first N well **241** and the second N well **242** and connected to a first (e.g., anode) collector terminal and/or electrode. The second isolation P well **272** is between the second N well **242** and the third N well **243** and connected to the first collector terminal and/or electrode. The third isolation P well **273** is between the third N well **243** and the fourth N well **244** and connected to the base terminal and/or electrode. The fourth isolation P well **274** is between the fourth N well **244** and the fifth N well **245** and connected to a second (e.g., cathode) collector terminal and/or electrode. The fifth isolation P well **275** is between the fifth N well **245** and the sixth N well **246** and connected to the second collector terminal and/or electrode.

(53) The first isolation deep well **281** to the fifth isolation deep well **285** are in the epitaxial layer **210**, and under the first isolation P well **271** to the fifth isolation P well **275**, respectively. The first isolation deep well **281** is under the first isolation P well **271**, and connected to a lower portion of the first isolation P well **271**. The second isolation deep well **282** is under the second isolation P well **272**, and connected to a lower portion of the second isolation P well **272**. The third isolation deep well **283** is under the third isolation P well **273**, and connected to a lower portion of the third isolation P well **273**. The fourth isolation deep well **284** is under the fourth isolation P well **274**, and connected to a lower portion of the fourth isolation P well **274**. The fifth isolation deep well **285** is under the fifth isolation P well **275**, and connected to a lower portion of the fifth isolation P well **275**.

(54) The third N well **243**, the third isolation P well **273**, and the fourth N well **244** connected to the base terminal and/or electrode in the center of the drawing constitute an isolation region **290**. Herein, the isolation region **290** serves as a region for isolation between an anode on the left side of the drawing and a cathode on the right side of the drawing. The isolation region **290** includes the third doped layer **253** and the second connection N well **262** connected to the third N well **243**, the third isolation deep well **283** connected to the third isolation P well **273**, the fourth doped layer **254** and the third connection N well **263** connected to the fourth N well **244**, and part of the buried layer **220**.

(55) Herein, the P-sub terminal and/or electrode and the collector terminals and/or electrodes on the left side of the drawing constitute the anode, and the collector terminals and/or electrodes on the right side of the drawing constitute the cathode. The device **200** may function as a type or kind of bidirectional diode, in which only voltages greater than a first positive threshold and voltages less than a second negative threshold pass through the device.

(56) The P well **230**, the first isolation P well **271**, the first isolation deep well **281**, the second isolation P well **272**, and the second isolation deep well **282** constitute an anode region. Herein, the

anode region may include the first N well **241**, the first doped layer **251**, the first connection N well **261**, the second N well **242**, and the second doped layer **252**.

(57) The fourth isolation P well **274**, the fourth isolation deep well **284**, the fifth isolation P well **275**, and the fifth isolation deep well **285** constitute a cathode region. Herein, the cathode region may include the fifth N well **245**, the fifth doped layer **255**, the sixth N well **246**, the sixth doped layer **256**, and the fourth connection N well **264**.

(58) The isolation region **290** constitutes an isolation structure having a very large BVcex between the anode region and the cathode region. That is, the BVcex of the isolation region **290** has a value greater than the breakdown voltages (BV2 @ cathode positive and BV1 @ cathode positive) when the cathode is subjected to positive zapping. Herein, the breakdown voltage BV2 @ cathode positive has a value greater than the breakdown voltage BV1 @ cathode positive, and has the relation below:

$BV_{cex} > BV2 @ \text{cathode positive} > BV1 @ \text{cathode positive}$

(59) In addition, the BVcex of the isolation region **290** has a value greater than breakdown voltages (BV3 @ cathode negative and BV4 @ cathode negative) when the cathode is subjected to negative zapping. Herein, the breakdown voltage BV4 @ cathode negative has a value greater than the breakdown voltage BV3 @ cathode negative, and has the relation below:

$BV_{cex} > BV4 @ \text{cathode negative} > BV3 @ \text{cathode negative}$

(60) In addition, the isolation P well **230** and the base (that is, a PNP device) in the anode region or the cathode region may comprise a multi-finger device or have a multi-finger layout.

(61) Accordingly, the bidirectional ESD protection device **200** according to the second exemplary embodiment of the present disclosure can secure the same breakdown voltages (BVs) in positive zapping and negative zapping when the PNP transistors in the anode region and the cathode region are the same.

(62) FIGS. **6** and **7** are diagrams illustrating a bidirectional ESD protection device according to a third exemplary embodiment of the present disclosure.

(63) Referring to FIGS. **6** and **7**, the bidirectional ESD protection device **300** according to the third exemplary embodiment of the present disclosure includes an epitaxial layer **310**, a buried layer **320**, a P well **330**, a first N well **341**, a second N well **342**, a third N well **343**, a fourth N well **344**, a fifth N well **345**, a first doped layer **351**, a second doped layer **352**, a third doped layer **353**, a fourth doped layer **354**, a fifth doped layer **355**, a first connection N well **361**, a second connection N well **362**, a third connection N well **363**, a first isolation P well **371**, a second isolation P well **372**, a third isolation P well **373**, a fourth isolation P well **374**, a first isolation deep well **381**, a second isolation deep well **382**, a third isolation deep well **383**, and a fourth isolation deep well **384**.

(64) The epitaxial layer **310** is on one surface of a semiconductor substrate **395**. The epitaxial layer **310** has the first conductivity type, and may comprise elemental silicon and be formed by an epitaxial growth process. For example, the epitaxial layer **310** may further comprise one or more dopants and be doped by adding the dopant(s) during the epitaxial growth process. The epitaxial layer may also be doped by ion implantation after the epitaxial layer is formed.

(65) The buried layer **320** is in the epitaxial layer **310** and having a predetermined width. The buried layer **320** is close to a lower surface of the epitaxial layer **310**, inside the epitaxial layer **310**. The buried layer **320** is connected to lower surfaces of the first connection N well **361** to the fourth connection N well **364**. The buried layer **320** comprises antimony, a material of the second conductivity type, for example.

(66) The P well **330** is in the epitaxial layer **310**, and at or adjacent to an upper surface of the epitaxial layer **310**. The P well **330** may be at one outermost side or border of the device **300** (e.g., close to the left side of the epitaxial layer **310** in the drawing). The P well **330** may be formed by injecting high-concentration first-conductivity-type impurities into the surface of the epitaxial layer **310** through a photolithographically-patterned mask. The P well **330** is connected to (e.g., in ohmic

contact with) a P-sub terminal and/or electrode.

(67) The first N well **341** to the fifth N well **345** are in the epitaxial layer **310**, and at or adjacent to the upper surface of the epitaxial layer **310**. The first N well **341** to the fifth N well **345** are between the right side of the epitaxial layer **310** and the P well **330** in the drawing, and spaced apart from each other by a predetermined distance. Herein, the third N well **343** in the center of the drawing has a larger area than the first N well **341**, the second N well **342**, the fourth N well **344**, and the fifth N well **345**.

(68) The first N well **341** is adjacent to the P well **330** and connected to an M1 terminal and/or electrode.

(69) The second N well **342** is adjacent to the first N well **341** and connected to an emitter terminal and/or electrode. That is, a P+ region adjacent to the second N well **342** is connected to the anode collector terminal and/or electrode, and the second N well **342** is connected to the anode emitter terminal and/or electrode.

(70) The third N well **343** is adjacent to the second N well **342** and connected to a base terminal and/or electrode.

(71) The fourth N well **344** is adjacent to the third N well **343** and connected to an emitter terminal and/or electrode. That is, a P+ region adjacent to the fourth N well **344** is connected to the cathode collector terminal and/or electrode, the fourth N well **344** is connected to the cathode emitter terminal and/or electrode.

(72) The fifth N well **345** is at or adjacent to the fourth N well **344** and connected to a base terminal and/or electrode. Thus, the second N well **342** is between the first N well **341** and the third N well **343**, the third N well **343** is between the second N well **342** and the fourth N well **344**, and the fourth N well **344** is between the third N well **343** and the fifth N well **345**.

(73) The first doped layer **351** to the fifth doped layer **355** are in the epitaxial layer **310**, and under the first N well **341** and the fifth N well **345**, respectively.

(74) The first doped layer **351** to the fifth doped layer **355** may overlap with parts of the first N well **341** to the fifth N well **345**, respectively. That is, the first doped layer **351** is under the first N well **341**, and may overlap with part of the first N well **341**. The second doped layer **352** is under the second N well **342**, and may overlap with part of the second N well **342**. The third doped layer **353** is under the third N well **343**, and may overlap with part of the third N well **343**. The fourth doped layer **354** is under the fourth N well **344**, and may overlap with part of the fourth N well **344**. The fifth doped layer **355** is under the fifth N well **345**, and may overlap with part of the fifth N well **345**.

(75) The first connection N well **361** to the third connection N well **363** are in the epitaxial layer **310**. The first connection N well **361** to the third connection N well **363** connect the buried layer **320** to the first doped layer **351**, the third doped layer **353**, and the fifth doped layer **355** or lower surfaces thereof. That is, in the drawing, an upper portion of the first connection N well **361** is connected to the first doped layer **351**, and a lower portion of the first connection N well **361** is connected to the buried layer **320**. In the drawing, an upper portion of the second connection N well **362** is connected to the third doped layer **353**, and a lower portion of the second connection N well **362** is connected to the buried layer **320**. In the drawing, an upper portion of the third connection N well **363** is connected to the fifth doped layer **355**, and a lower portion of the third connection N well **363** is connected to the buried layer **320**.

(76) The first isolation P well **371** to the fourth isolation P well **374** are in the epitaxial layer **310**, and at or adjacent to the upper surface of the epitaxial layer **310**. The first isolation P well **371** to the fourth isolation P well **374** are spaced apart from each other by a predetermined distance, and are respectively between adjacent ones of the N wells.

(77) The first isolation P well **371** is between the first N well **341** and the second N well **342** and connected to a first (e.g., anode) collector terminal and/or electrode. The second isolation P well **372** is between the second N well **342** and the third N well **343** and connected to the first collector

terminal and/or electrode. The third isolation P well **373** is between the third N well **343** and the fourth N well **344** and connected to a second (e.g., cathode) collector terminal and/or electrode. The fourth isolation P well **374** is between the fourth N well **344** and the fifth N well **345** and connected to the second collector terminal and/or electrode.

(78) The first isolation deep well **381** to the fourth isolation deep well **384** are formed in the epitaxial layer **310**, and under the first isolation P well **371** to the fourth isolation P well **374**. The first isolation deep well **381** is under the first isolation P well **371**, and connected to a lower portion of the first isolation P well **371**. The second isolation deep well **382** is under the second isolation P well **372**, and connected to a lower portion of the second isolation P well **372**. The third isolation deep well **383** is under the third isolation P well **373**, and connected to a lower portion of the third isolation P well **373**. The fourth isolation deep well **384** is under the fourth isolation P well **374**, and connected to a lower portion of the fourth isolation P well **374**.

(79) The third N well **343**, the third doped layer **353**, and the second connection N well **362** connected to the base terminal and/or electrode positioned in the center of the drawing constitute an isolation region **390**. Herein, the isolation region **390** serves as a region for isolation between an anode on the left side of the drawing and a cathode on the right side of the drawing.

(80) Herein, the P-sub terminal and/or electrode and the collector terminals and/or electrode on the left side of the drawing constitute the anode, and the collector terminals and/or electrode on the right side of the drawing constitute the cathode. The device **300** may function as a type or kind of bidirectional diode, in which only voltages greater than a first positive threshold and voltages less than a second negative threshold pass through the device.

(81) The P well **330**, the first isolation P well **371**, the first isolation deep well **381**, the second isolation P well **372**, and the second isolation deep well **382** constitute an anode region. Herein, the anode region may include the first N well **341**, the first doped layer **351**, the first connection N well **361**, the second N well **342**, and the second doped layer **352**.

(82) The third isolation P well **373**, the third isolation deep well **383**, the fourth isolation P well **374**, and the fourth isolation deep well **384** constitute a cathode region. Herein, the cathode region may include the fourth N well **344**, the fourth doped layer **354**, the fifth N well **345**, the third doped layer **353**, and the third connection N well **363**.

(83) The isolation region **390** constitutes an isolation structure having a very large BV_{cex} between the anode region and the cathode region. That is, the BV_{cex} of the isolation region **390** has a value greater than the breakdown voltages (BV2 @ cathode positive and BV1 @ cathode positive) when the cathode is subjected to positive zapping. Herein, the breakdown voltage BV2 @ cathode positive has a value greater than the breakdown voltage BV1 @ cathode positive, and has the relation below:

$BV_{cex} > BV2 @ \text{cathode positive} > BV1 @ \text{cathode positive}$

(84) In addition, the BV_{cex} of the isolation region **390** has a value greater than breakdown voltages (BV3 @ cathode negative and BV4 @ cathode negative) when the cathode is subjected to negative zapping. Herein, the breakdown voltage BV4 @ cathode negative has a value greater than the breakdown voltage BV3 @ cathode negative, and has the relation below:

$BV_{cex} > BV4 @ \text{cathode negative} > BV3 @ \text{cathode negative}$

(85) In addition, the isolation P well **330** and the base (that is, a PNP device) in the anode region or the cathode region may comprise a multi-finger device or have a multi-finger layout.

(86) Accordingly, the bidirectional ESD protection device **300** according to the third exemplary embodiment of the present disclosure can secure the same breakdown voltages (BVs) in positive zapping and negative zapping when the PNP transistors in the anode region and the cathode region are the same.

(87) FIG. **8** is a diagram illustrating comparisons between the bidirectional ESD protection devices according to the exemplary embodiments of the present disclosure and the conventional bidirectional ESD protection device.

(88) Referring to FIG. 8, the conventional bidirectional ESD protection device has an anode and a cathode identical in junction design, but a large breakdown voltage (BV) mismatch occurs (see the left-hand “Impact ionization rate” graph in FIG. 8).

(89) Conversely, when the bidirectional ESD protection devices **100**, **200**, and **300** according to the embodiments of the present disclosure have an anode and a cathode identical in junction design, there is almost no breakdown voltage (BV) mismatch (see the right-hand “Impact ionization rate” graph in FIG. 8), compared to the conventional bidirectional ESD protection device.

(90) Accordingly, the bidirectional ESD protection devices **100**, **200**, and **300** according to the embodiments of the present disclosure realize the same PNP structures in the anode region and the cathode region, thereby preventing a breakdown voltage mismatch.

(91) In addition, the bidirectional ESD protection devices **100**, **200**, and **300** according to the embodiments of the present disclosure can solve the problem in the related art, an unstable breakdown voltage, by providing a high-voltage region (e.g., of the bidirectional ESD protection devices) with a stable breakdown voltage.

(92) Although the exemplary embodiments of the present disclosure have been described, those skilled in the art will appreciate that various modifications, additions and substitutions are possible, without departing from the scope and spirit of the disclosure as disclosed in the accompanying claims.

Claims

1. A bidirectional ESD protection device, comprising: a P-type epitaxial layer; an N-type buried layer in the epitaxial layer; a P well at or adjacent to an upper surface of the epitaxial layer, a plurality of N wells at or adjacent to the upper surface of the epitaxial layer and spaced apart from each other; a plurality of connection N wells connected to the buried layer, in the epitaxial layer, a plurality of isolation P wells at or adjacent to the upper surface of the epitaxial layer and respectively between adjacent ones of the plurality of N wells; and an anode region, a cathode region, and an isolation region between the anode region and the cathode region, wherein the isolation region comprises two adjacent ones of the plurality of N wells and the isolation P well between the two adjacent ones of the plurality of N wells, and the anode region and the cathode region comprise remaining ones of the plurality of N wells and remaining ones of the plurality of isolation P wells.
2. The bidirectional ESD protection device of claim 1, wherein the plurality of N wells comprise: a first N well at or adjacent to the upper surface of the epitaxial layer and spaced apart from the P well; a second N well at or adjacent to the upper surface of the epitaxial layer and spaced apart from the first N well; a third N well at or adjacent to the upper surface of the epitaxial layer and spaced apart from the second N well; a fourth N well at or adjacent to the upper surface of the epitaxial layer and spaced apart from the third N well; a fifth N well at or adjacent to the upper surface of the epitaxial layer and spaced apart from the fourth N well; and a sixth N well at or adjacent to the upper surface of the epitaxial layer and spaced apart from the fifth N well.
3. The bidirectional ESD protection device of claim 2, further comprising a plurality of doped layers under the N wells, respectively, in the epitaxial layer, wherein the plurality of doped layers comprise: a first doped layer at least partially surrounding the first N well, in the epitaxial layer, a second doped layer at least partially surrounding the second N well, in the epitaxial layer; a third doped layer at least partially surrounding the third N well, in the epitaxial layer, a fourth doped layer at least partially surrounding the fourth N well, in the epitaxial layer, a fifth doped layer at least partially surrounding the fifth N well, in the epitaxial layer, and a sixth doped layer at least partially surrounding the sixth N well, in the epitaxial layer.
4. The bidirectional ESD protection device of claim 3, wherein the plurality of connection N wells comprise: a first connection N well connecting the first doped layer with the buried layer, in the

epitaxial layer; a second connection N well connecting the second doped layer with the buried layer, in the epitaxial layer; a third connection N well connecting the third doped layer with the buried layer, in the epitaxial layer; a fourth connection N well connecting the fourth doped layer with the buried layer, in the epitaxial layer; a fifth connection N well connecting the fifth doped layer with the buried layer, in the epitaxial layer; and a sixth connection N well connecting the sixth doped layer with the buried layer, in the epitaxial layer.

5. The bidirectional ESD protection device of claim 2, wherein the plurality of isolation P wells comprise: a first isolation P well at or adjacent to the upper surface of the epitaxial layer, and between the first N well and the second N well; a second isolation P well at or adjacent to the upper surface of the epitaxial layer, and between the second N well and the third N well; a third isolation P well at or adjacent to the upper surface of the epitaxial layer, and between the third N well and the fourth N well; a fourth isolation P well at or adjacent to the upper surface of the epitaxial layer, and between the fourth N well and the fifth N well; and a fifth isolation P well at or adjacent to the upper surface of the epitaxial layer, and between the fifth N well and the sixth N well.

6. The bidirectional ESD protection device of claim 5, further comprising a plurality of isolation deep wells connected to lower portions of the isolation P wells, respectively, in the epitaxial layer, wherein the plurality of isolation deep wells comprise: a first isolation deep well connected to the first isolation P well, in the epitaxial layer; a second isolation deep well connected to the second isolation P well, in the epitaxial layer, a third isolation deep well connected to the third isolation P well, in the epitaxial layer; a fourth isolation deep well connected to the fourth isolation P well, in the epitaxial layer, and a fifth isolation deep well connected to the fifth isolation P well, in the epitaxial layer.

7. The bidirectional ESD protection device of claim 5, wherein the isolation region comprises the third N well, the fourth N well, and the third isolation P well, and the anode region comprises the first N well, the second N well, the first isolation P well, and the second isolation P well on one side of the isolation region, and the cathode region comprises the fifth N well, the sixth N well, the fourth isolation P well, and the fifth isolation P well on an opposite side of the isolation region.

8. A bidirectional ESD protection device, comprising: a P-type epitaxial layer; an N-type buried layer in the epitaxial layer; a P well at or adjacent to an upper surface of the epitaxial layer, a plurality of N wells at or adjacent to the upper surface of the epitaxial layer and spaced apart from each other; a plurality of connection N wells connected to the buried layer, in the epitaxial layer, and a plurality of isolation P wells at or adjacent to the upper surface of the epitaxial layer and respectively between adjacent ones of the plurality of N wells; and an anode region, a cathode region, and an isolation region between the anode region and the cathode region, wherein the isolation region comprises two adjacent ones of the plurality of N wells and the isolation P well between the two adjacent ones of the two N wells, and the anode region and the cathode region face each other with the isolation region in a center, and the anode region and the cathode region are on opposite sides of the isolation region.

9. The bidirectional ESD protection device of claim 8, wherein the plurality of N wells comprise: a first N well at or adjacent to the upper surface of the epitaxial layer and spaced apart from the P well; a second N well at or adjacent to the upper surface of the epitaxial layer and spaced apart from the first N well; a third N well at or adjacent to the upper surface of the epitaxial layer and spaced apart from the second N well; a fourth N well at or adjacent to the upper surface of the epitaxial layer and spaced apart from the third N well; a fifth N well at or adjacent to the upper surface of the epitaxial layer and spaced apart from the fourth N well; and a sixth N well at or adjacent to the upper surface of the epitaxial layer and spaced apart from the fifth N well.

10. The bidirectional ESD protection device of claim 9, further comprising a plurality of doped layers under the N wells, respectively, in the epitaxial layer, wherein the plurality of doped layers comprise: a first doped layer at least partially surrounding the first N well, in the epitaxial layer, a second doped layer at least partially surrounding the second N well, in the epitaxial layer; a third

doped layer at least partially surrounding the third N well, in the epitaxial layer, a fourth doped layer at least partially surrounding the fourth N well, in the epitaxial layer, a fifth doped layer at least partially surrounding the fifth N well, in the epitaxial layer, and a sixth doped layer at least partially surrounding the sixth N well, in the epitaxial layer.

11. The bidirectional ESD protection device of claim 10, wherein the plurality of connection N wells comprise: a first connection N well under the first doped layer connecting the first doped layer with the buried layer, in the epitaxial layer; a second connection N well under the third doped layer connecting the third doped layer with the buried layer, in the epitaxial layer; a third connection N well under the fourth doped layer connecting the fourth doped layer with the buried layer, in the epitaxial layer; and a fourth connection N well under the sixth doped layer connecting the sixth doped layer with the buried layer, in the epitaxial layer.

12. The bidirectional ESD protection device of claim 9, wherein the plurality of isolation P wells comprise: a first isolation P well at or adjacent to the upper surface of the epitaxial layer, and between the first N well and the second N well; a second isolation P well at or adjacent to the upper surface of the epitaxial layer, and between the second N well and the third N well; a third isolation P well at or adjacent to the upper surface of the epitaxial layer, and between the third N well and the fourth N well; a fourth isolation P well at or adjacent to the upper surface of the epitaxial layer, and between the fourth N well and the fifth N well; and a fifth isolation P well at or adjacent to the upper surface of the epitaxial layer, and between the fifth N well and the sixth N well.

13. The bidirectional ESD protection device of claim 12, further comprising a plurality of isolation deep wells connected to lower portions of the isolation P wells, respectively, in the epitaxial layer, wherein the plurality of isolation deep wells comprises: a first isolation deep well connected to the first isolation P well, in the epitaxial layer; a second isolation deep well connected to the second isolation P well, in the epitaxial layer, a third isolation deep well connected to the third isolation P well, in the epitaxial layer; a fourth isolation deep well connected to the fourth isolation P well, in the epitaxial layer, and a fifth isolation deep well connected to the fifth isolation P well, in the epitaxial layer.

14. The bidirectional ESD protection device of claim 12, wherein the isolation region comprises the third N well, the fourth N well, and the third isolation P well, the anode region comprises the first N well, the second N well, the first isolation P well, and the second isolation P well, the cathode region comprises the fifth N well, the sixth N well, the fourth isolation P well, and the fifth isolation P well, the anode region and the cathode region are on opposite sides of the isolation region, and the first N well, the second N well, the first isolation P well, and the second isolation P well face the fifth N well, the sixth N well, the fourth isolation P well, and the fifth isolation P well, with the isolation region in the center.

15. A bidirectional ESD protection device, comprising: a P-type epitaxial layer; an N-type buried layer in the epitaxial layer; a P well at or adjacent to an upper surface of the epitaxial layer, a plurality of N wells at or adjacent to the upper surface of the epitaxial layer and spaced apart from each other; a plurality of connection N wells connected to the buried layer, in the epitaxial layer, and a plurality of isolation P wells at or adjacent to the upper surface of the epitaxial layer and respectively between each of the plurality of N wells; and an anode region, a cathode region, and an isolation region between the anode region and the cathode region, wherein the isolation region comprises one of the plurality of N wells, and the anode region and the cathode region comprise remaining ones of the plurality of N wells and the plurality of isolation P wells, the remaining ones of the plurality of N wells and the plurality of isolation P wells face each other with the isolation region in a center, and the anode region and the cathode region are on opposite sides of the isolation region.

16. The bidirectional ESD protection device of claim 15, wherein the plurality of N wells comprise: a first N well at or adjacent to the upper surface of the epitaxial layer and spaced apart from the P well; a second N well at or adjacent to the upper surface of the epitaxial layer and spaced apart

from the first N well; a third N well at or adjacent to the upper surface of the epitaxial layer and spaced apart from the second N well; a fourth N well at or adjacent to the upper surface of the epitaxial layer and spaced apart from the third N well; and a fifth N well at or adjacent to the upper surface of the epitaxial layer and spaced apart from the fourth N well, and the plurality of isolation P wells comprise: a first isolation P well at or adjacent to the upper surface of the epitaxial layer, and between the first N well and the second N well; a second isolation P well at or adjacent to the upper surface of the epitaxial layer, and between the second N well and the third N well; a third isolation P well at or adjacent to the upper surface of the epitaxial layer, and between the third N well and the fourth N well; and a fourth isolation P well at or adjacent to the upper surface of the epitaxial layer, and between the fourth N well and the fifth N well.

17. The bidirectional ESD protection device of claim 16, further comprising a plurality of doped layers comprising: a first doped layer at least partially surrounding the first N well, in the epitaxial layer; a second doped layer at least partially surrounding the second N well, in the epitaxial layer, a third doped layer at least partially surrounding the third N well, in the epitaxial layer, a fourth doped layer at least partially surrounding the fourth N well, in the epitaxial layer, and a fifth doped layer at least partially surrounding the fifth N well, in the epitaxial layer.

18. The bidirectional ESD protection device of claim 17, wherein the plurality of connection N wells comprise: a first connection N well connecting the first doped layer with the buried layer, in the epitaxial layer; a second connection N well connecting the third doped layer with the buried layer, in the epitaxial layer; and a third connection N well connecting the fifth doped layer with the buried layer, in the epitaxial layer.

19. The bidirectional ESD protection device of claim 16, further comprising a plurality of isolation deep wells connected to lower portions of the isolation P wells, respectively, in the epitaxial layer, wherein the plurality of isolation deep wells comprise: a first isolation deep well connected to the first isolation P well under the first isolation P well, in the epitaxial layer, a second isolation deep well connected to the second isolation P well under the second isolation P well, in the epitaxial layer, a third isolation deep well connected to the third isolation P well under the third isolation P well, in the epitaxial layer, and a fourth isolation deep well connected to the fourth isolation P well under the fourth isolation P well, in the epitaxial layer.

20. The bidirectional ESD protection device of claim 16, wherein the isolation region comprises the third N well, the anode region comprises the first N well, the second N well, the first isolation P well, and the second isolation P well, the cathode region comprises the fourth N well, the fifth N well, the third isolation P well, and the fourth isolation P well, and the first N well, the second N well, the first isolation P well, and the second isolation P well face the fourth N well, the fifth N well, the third isolation P well, and the fourth isolation P well, with the isolation region in the center.
